

Geometrically-informed revision to the intersection-point-height measure of human standing

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Abstract

Our understanding of the neuromuscular control underlying human standing balance remains limited. Because current technology cannot simultaneously monitor all internal biological signals, computational models are essential for inferring possible control mechanisms from experimental data. However, to accurately describe human standing, these models require quantitative experimental validation. One valuable measure for this purpose is the intersection-point height, which quantifies the co-variation between center-of-pressure displacements and foot-ground force orientation during standing. Widely applied across diverse populations during unperturbed standing, this measure has been computationally shown to capture aspects of neuromuscular control, highlighting its potential for studying balance mechanisms in healthy versus impaired individuals. Here, we propose a geometrically-informed modification to the intersection-point-height measure, inspired by its original geometric interpretation. We detail how to compute this revised measure using both experimental data and computational models, and provide open-source example code in MATLAB to facilitate its application.

1 Introduction

Standing balance ability promotes mobility and independent living. However, despite the importance of standing balance and its apparent simplicity, our understanding of the neuromusculoskeletal control of human standing balance is still limited. The challenge of studying postural control is due in part to the complexity of the biological and mechanical components of the system: the human body has numerous degrees of freedom that have to be coordinated simultaneously, and there are multiple sensors and actuators that all communicate in the control of balance. With current technology, it is impossible to monitor all the internal biological signals simultaneously to fully characterize the system function.

One strategy to reveal component functions of the human neuromuscular system is to use computational models and simulations. Models can identify the apparent balance controller that best describes experimental data, from which we can infer possible mechanisms. For this approach to adequately describe human standing, the models must be validated with quantitative experimental observations. Observations that distinguish between healthy and impaired balance, and between types or levels of impairment, would be particularly useful, as they can help validate models for specific impairments. These models may then be used to diagnose the cause of the impairment and guide patient treatment or assistance.

One recently developed measure is the intersection-point-height measure that quantifies the co-variation between the center-of-pressure (CoP) displacements and foot-ground-interaction force orientation in a vertical plane of interest (e.g. frontal plane, sagittal plane) [1]. The intersection-point-height measure has been applied to various standing tasks and has successfully differentiated standing data from various surfaces [2, 3], age groups [4], as well as post-stroke versus unimpaired [5, 6]. Computational models have also been applied to show that the intersection-point height captures the role of the neuromuscular balance controller and not just mechanical properties [7, 8]. The intersection-point height appears to be a useful measure to quantify the closed-loop balance dynamics to better understand balance mechanisms.

Here, we describe a geometrically-informed modification to the intersection-point-height measure that takes inspiration from the original geometric interpretation of the measure. The intersection point was understood to be the spatial point that minimized the orthogonal distances from all lines-of-action of foot forces. If the vertical component of the foot force is much larger than the horizontal component (which is true for

quiet standing), then the height of this geometrically-defined intersection point can be approximated as the slope of the regression line of CoP displacements versus foot-force orientation. This quantity remains closely related to the originally defined intersection-point-height measure [1], capturing the closed-loop standing balance dynamics, including the effects of the apparent controller and sensorimotor noise [8], and thus is favorable for application in controller identification and model validation. MATLAB code for this newly defined intersection-point height is available online at https://github.com/rikasd/zIP_poststroke, where the method has been applied to analyze quiet-standing data from survivors of stroke and develop computational models describing the observed balance behavior [6].

The remainder of the paper is organized as follows: in Section 2, the geometric intersection point is derived; and in Section 3, a revision to the intersection-point-height measure of standing is proposed based on the geometric definition. The latter half of Section 3 also includes a detailed account of how the revised measure can be computed from human standing data and computational models. Section 4 summarizes the contributions of the new intersection-point-height measure and discusses its strengths and limitations. Section 5 closes with concluding remarks.

2 Geometrically-defined intersection point

The intersection point (IP) of the foot-force during quiet standing is geometrically defined as the point in a vertical plane (frontal or sagittal) from which all the foot-force lines of action minimally deviate. Consider Figure 1: point P at coordinates (x_P, z_P) is the geometric intersection point IP_{gm} of the foot-forces $\{F_1, F_2, F_3\}$ when the sum of squares of the orthogonal distance between the force lines of action and P is minimized, i.e.,

$$(x_{IP_{gm}}, z_{IP_{gm}}) = \arg \min_{(x_P, z_P)} \sum_{i=1}^N \delta_i^2, \quad (1)$$

where $(x_{IP_{gm}}, z_{IP_{gm}})$ are the coordinates of IP_{gm} , δ_i is the orthogonal distance between force F_i line of action and point P , and $N = 3$ is the number of foot-force data points.

We can solve for the optimization problem 1 by re-writing δ_i in terms of the coordinates (x_P, z_P) and the foot-force data $\{x_{CoP_i}, F_{x_i}, F_{z_i}\}$, taking the derivative of 1 with respect to x_P and z_P , and setting them to zero to solve for (x_P, z_P) .

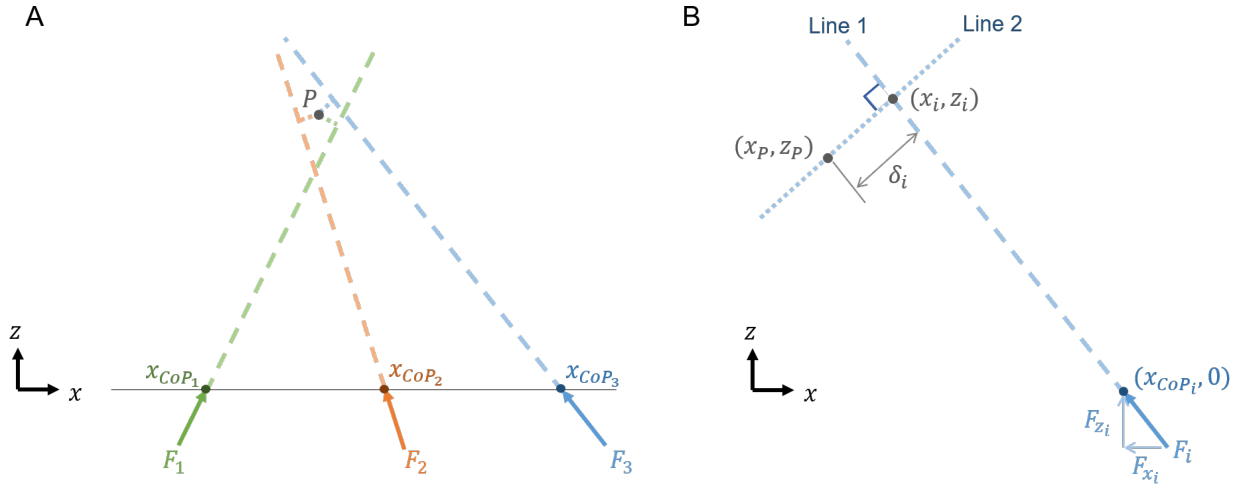


Figure 1: Geometric definition of the intersection point. (A) Point P is the intersection point (IP) of the foot-forces $\{F_1, F_2, F_3\}$ when the sum of squares of the orthogonal distance between the force lines-of-action and P is minimized. (B) The orthogonal distance between the force line of action F_i and point P at coordinates (x_P, z_P) is shown as δ_i .

Let (x_i, z_i) be the coordinates of the point along the force F_i line of action (labeled as line 1 in Figure 1B) that is closest to point P . The orthogonal distance squared can be written in terms of (x_P, z_P) and (x_i, z_i) :

$$\delta_i^2 = (x_i - x_P)^2 + (z_i - z_P)^2. \quad (2)$$

Consider line 2 joining (x_i, z_i) and (x_P, z_P) . Line 2 is perpendicular to line 1, and is oriented along the vector $\vec{v}_2 = [F_{z_i}, -F_{x_i}]$. The equation of line 2 can be written in terms of $\vec{v}_2$ and (x_P, z_P) as:

$$\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} x_P \\ z_P \end{bmatrix} + s \begin{bmatrix} F_{z_i} \\ -F_{x_i} \end{bmatrix}, \quad (3)$$

where s is a scalar. Since the point (x_i, z_i) lies along line 2, its coordinates can be given by:

$$x_i = x_P + F_{z_i} s_i \quad (4)$$

$$z_i = z_P - F_{x_i} s_i, \quad (5)$$

for some scalar $s = s_i$. Substituting these expressions for x_i and z_i into equation 2, one obtains:

$$\delta_i^2 = (F_{z_i} s_i)^2 + (F_{x_i} s_i)^2 = (F_{z_i}^2 + F_{x_i}^2) s_i^2. \quad (6)$$

To solve for the unknown scalar s_i , we consider the equation of line 1, which is oriented along vector $\vec{v}_1 = [F_{x_i}, F_{z_i}]$ and intersects the point $(x_{CoP_i}, 0)$:

$$\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} x_{CoP_i} \\ 0 \end{bmatrix} + r \begin{bmatrix} F_{x_i} \\ F_{z_i} \end{bmatrix}, \quad (7)$$

where r is a scalar. Since the point (x_i, z_i) also lies along line 1, we can obtain another set of equations in terms of some scalar $r = r_i$:

$$x_i = x_{CoP_i} + F_{x_i} r_i \quad (8)$$

$$z_i = F_{z_i} r_i. \quad (9)$$

We can equate equations 8 and 4, 9 and 5, then use the two equations to solve for s_i in terms of (x_P, z_P) and $\{x_{CoP_i}, F_{x_i}, F_{z_i}\}$:

$$\begin{aligned} x_{CoP_i} + F_{x_i} r_i &= x_P + F_{z_i} s_i \\ F_{z_i} r_i &= z_P - F_{x_i} s_i, \end{aligned}$$

$$\begin{aligned} x_{CoP_i} F_{z_i} + F_{x_i} F_{z_i} r_i &= F_{z_i} x_P + F_{z_i}^2 s_i \\ -F_{x_i} F_{z_i} r_i &= -F_{x_i} z_P + F_{x_i}^2 s_i, \end{aligned}$$

$$x_{CoP_i} F_{z_i} = F_{z_i} x_P - F_{x_i} z_P + (F_{z_i}^2 + F_{x_i}^2) s_i,$$

$$s_i = \frac{1}{F_{z_i}^2 + F_{x_i}^2} \left(F_{z_i} (x_{CoP_i} - x_P) + F_{x_i} z_P \right).$$

Plugging this expression for s_i into equation 6 gives

$$\begin{aligned} \delta_i^2 &= (F_{z_i}^2 + F_{x_i}^2) \left(\frac{1}{F_{z_i}^2 + F_{x_i}^2} \right)^2 \left(F_{z_i} (x_{CoP_i} - x_P) + F_{x_i} z_P \right)^2 \\ &= \frac{1}{F_{z_i}^2 + F_{x_i}^2} \left(F_{z_i} (x_{CoP_i} - x_P) + F_{x_i} z_P \right)^2. \end{aligned}$$

Thus, the optimization problem of 1 becomes

$$(x_{\text{IPgm}}, z_{\text{IPgm}}) = \arg \min_{(x_P, z_P)} \sum_{i=1}^N \frac{1}{\|F_i\|^2} \left(F_{z_i}(x_{CoP_i} - x_P) + F_{x_i}z_P \right)^2, \quad (10)$$

where $\|F_i\|^2 = F_{z_i}^2 + F_{x_i}^2$.

The point (x_P, z_P) that minimizes the sum of squared distances must satisfy

$$\frac{d}{dx_P} \sum_{i=1}^N \delta_i^2 = 0 \quad \text{and} \quad \frac{d}{dz_P} \sum_{i=1}^N \delta_i^2 = 0.$$

Computing the derivatives:

$$\begin{aligned} \frac{d}{dx_P} \sum_{i=1}^N \frac{1}{\|F_i\|^2} \left(F_{z_i}(x_{CoP_i} - x_P) + F_{x_i}z_P \right)^2 &= 0 \\ \frac{d}{dz_P} \sum_{i=1}^N \frac{1}{\|F_i\|^2} \left(F_{z_i}(x_{CoP_i} - x_P) + F_{x_i}z_P \right)^2 &= 0, \end{aligned}$$

$$\begin{aligned} \sum_{i=1}^N \frac{-2F_{z_i}}{\|F_i\|^2} \left(F_{z_i}(x_{CoP_i} - x_P) + F_{x_i}z_P \right) &= 0 \\ \sum_{i=1}^N \frac{2F_{x_i}}{\|F_i\|^2} \left(F_{z_i}(x_{CoP_i} - x_P) + F_{x_i}z_P \right) &= 0, \end{aligned}$$

and expanding the summation:

$$\begin{aligned} - \sum_{i=1}^N \frac{F_{z_i}^2 x_{CoP_i}}{\|F_i\|^2} + x_P \sum_{i=1}^N \frac{F_{z_i}^2}{\|F_i\|^2} - z_P \sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{\|F_i\|^2} &= 0 \\ \sum_{i=1}^N \frac{F_{z_i} F_{x_i} x_{CoP_i}}{\|F_i\|^2} - x_P \sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{\|F_i\|^2} + z_P \sum_{i=1}^N \frac{F_{x_i}^2}{\|F_i\|^2} &= 0, \end{aligned}$$

we obtain:

$$\begin{bmatrix} \sum_{i=1}^N \frac{F_{z_i}^2}{\|F_i\|^2} & - \sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{\|F_i\|^2} \\ - \sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{\|F_i\|^2} & \sum_{i=1}^N \frac{F_{x_i}^2}{\|F_i\|^2} \end{bmatrix} \begin{bmatrix} x_P \\ z_P \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^N \frac{F_{z_i}^2 x_{CoP_i}}{\|F_i\|^2} \\ - \sum_{i=1}^N \frac{F_{z_i} F_{x_i} x_{CoP_i}}{\|F_i\|^2} \end{bmatrix}.$$

We can use the above matrix equation to solve for (x_P, z_P) , which correspond to the geometric intersection-point coordinates $(x_{\text{IPgm}}, z_{\text{IPgm}})$. Thus, we arrive at the following expression for the geometric intersection-point height:

$$z_{\text{IPgm}} = \frac{\sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{\|F_i\|^2} \sum_{i=1}^N \frac{F_{z_i}^2 x_{CoP_i}}{\|F_i\|^2} - \sum_{i=1}^N \frac{F_{z_i}^2}{\|F_i\|^2} \sum_{i=1}^N \frac{F_{z_i} F_{x_i} x_{CoP_i}}{\|F_i\|^2}}{\sum_{i=1}^N \frac{F_{z_i}^2}{\|F_i\|^2} \sum_{i=1}^N \frac{F_{x_i}^2}{\|F_i\|^2} - \left(\sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{\|F_i\|^2} \right)^2}. \quad (11)$$

3 Approximation of the intersection-point height via linear regression

During quiet stance, the vertical component of the foot-force dominates ($F_{z_i} \gg F_{x_i}$). Thus we can assume:

$$\|F_i\|^2 = F_{z_i}^2 + F_{x_i}^2 \approx F_{z_i}^2.$$

and expression 11 for the geometric intersection-point height simplifies to

$$z_{\text{IPgm}} \approx \frac{\sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{F_{z_i}^2} \sum_{i=1}^N \frac{F_{z_i}^2 x_{CoP_i}}{F_{z_i}^2} - \sum_{i=1}^N \frac{F_{z_i}^2}{F_{z_i}^2} \sum_{i=1}^N \frac{F_{z_i} F_{x_i} x_{CoP_i}}{F_{z_i}^2}}{\sum_{i=1}^N \frac{F_{z_i}^2}{F_{z_i}^2} \sum_{i=1}^N \frac{F_{x_i}^2}{F_{z_i}^2} - \left(\sum_{i=1}^N \frac{F_{z_i} F_{x_i}}{F_{z_i}^2} \right)^2},$$

$$\approx \frac{\sum_{i=1}^N \frac{F_{x_i}}{F_{z_i}} \sum_{i=1}^N x_{CoP_i} - N \sum_{i=1}^N \frac{F_{x_i} x_{CoP_i}}{F_{z_i}}}{N \sum_{i=1}^N \left(\frac{F_{x_i}}{F_{z_i}} \right)^2 - \left(\sum_{i=1}^N \frac{F_{x_i}}{F_{z_i}} \right)^2}.$$

Define $F_{\text{ratio},i} = F_{x_i}/F_{z_i}$. Substituting $F_{\text{ratio},i}$ into the equation,

$$z_{\text{IPgm}} \approx \frac{\sum_{i=1}^N F_{\text{ratio},i} \sum_{i=1}^N x_{CoP_i} - N \sum_{i=1}^N F_{\text{ratio},i} x_{CoP_i}}{N \sum_{i=1}^N F_{\text{ratio},i}^2 - \left(\sum_{i=1}^N F_{\text{ratio},i} \right)^2},$$

and re-arranging,

$$z_{\text{IPgm}} \approx - \left(\frac{\frac{1}{N} \sum_{i=1}^N F_{\text{ratio},i} x_{CoP_i} - \frac{1}{N} \sum_{i=1}^N F_{\text{ratio},i} \frac{1}{N} \sum_{i=1}^N x_{CoP_i}}{\frac{1}{N} \sum_{i=1}^N F_{\text{ratio},i}^2 - \frac{1}{N} \sum_{i=1}^N F_{\text{ratio},i} \frac{1}{N} \sum_{i=1}^N F_{\text{ratio},i}} \right),$$

we recognize the numerator as the sample covariance of F_{ratio} and x_{CoP} , and the denominator as the sample variance of F_{ratio} . Thus, the geometric intersection-point height can be approximated by:

$$z_{\text{IPgm}} \approx - \frac{\text{cov}(F_{\text{ratio}}, x_{CoP})}{\text{var}(F_{\text{ratio}})}.$$

Since $F_{z_i} \gg F_{x_i}$ during quiet stance, the force ratio is approximately equal to the negative of the force orientation: $F_{\text{ratio}} \approx -\theta_F$. Thus, the geometric intersection-point height may also be given by:

$$z_{\text{IPgm}} \approx \frac{\text{cov}(\theta_F, x_{CoP})}{\text{var}(\theta_F)}.$$

The above expression for z_{IPgm} is equivalent to the slope of the regression line between x_{CoP} and θ_F . Thus, moving forward, **we refer to the approximate geometric intersection-point height as the linear-regression-based (LR-based) intersection-point height, denoted by $z_{\text{IP,LR}}$.**

The "goodness-of-fit" of the LR-based intersection point may be given by the square of the correlation coefficient, also known as the coefficient of determination:

$$R^2 = \frac{\text{cov}(\theta_F, x_{CoP})^2}{\text{var}(\theta_F) \text{var}(x_{CoP})}.$$

When $F_{z_i} \gg F_{x_i}$, the mean squared orthogonal distance between the force lines of action and the intersection point is related to R^2 by:

$$\frac{1}{N} \sum_{i=1}^N \delta_i^2 \approx \text{var}(x_{CoP}) (1 - R^2).$$

Indeed, if $R^2 = 1$ then $\sum_{i=1}^N \delta_i^2 \approx 0$; that is, when the linear regression perfectly describes the relationship between θ_F and x_{CoP} , then all force lines of action (almost exactly) intersect at height z_{IPgm} .

3.1 Frequency-dependent intersection-point-height measure of human standing

Using the geometrically derived LR-based intersection-point height $z_{\text{IP,LR}}$, here we define the intersection-point height at a given frequency as:

$$z_{\text{IP,LR}}(f) = \frac{\text{cov}\left(\theta_F(t; f, B_e), x_{\text{CoP}}(t; f, B_e)\right)}{\text{var}\left(\theta_F(t; f, B_e)\right)}, \quad (12)$$

where $\theta_F(t; f, B_e)$ and $x_{\text{CoP}}(t; f, B_e)$ denote $\theta_F(t)$ and $x_{\text{CoP}}(t)$ bandpass-filtered at center frequency f with bandwidth B_e . The goodness-of-fit of the IP height at each frequency can also be computed using the bandpass-filtered quantities:

$$R^2(f) = \frac{\text{cov}\left(\theta_F(t; f, B_e), x_{\text{CoP}}(t; f, B_e)\right)^2}{\text{var}\left(\theta_F(t; f, B_e)\right)\text{var}\left(x_{\text{CoP}}(t; f, B_e)\right)}.$$

As an alternative to bandpass-filtering time series data, the intersection-point height can also be computed directly in the frequency domain via spectral analysis. The covariance between θ_F and x_{CoP} bandpass-filtered at an infinitesimally narrow frequency band ($B_e \rightarrow 0$) about a central frequency f is equal to the real part of the one-sided cross-spectral density (also known as the "co-spectral density") between θ_F and x_{CoP} at that same frequency [9]:

$$\lim_{B_e \rightarrow 0} \text{cov}\left(\theta_F(t; f, B_e), x_{\text{CoP}}(t; f, B_e)\right) = \Re\left\{G_{\theta_F x_{\text{CoP}}}(f)\right\}.$$

Similarly, the variances of the bandpass-filtered θ_F and x_{CoP} , at the limit of infinitesimally narrow bandwidth $B_e \rightarrow 0$, are equal to their (real-valued) one-sided auto-spectral densities:

$$\lim_{B_e \rightarrow 0} \text{var}\left(\theta_F(t; f, B_e)\right) = G_{\theta_F \theta_F}(f),$$

$$\lim_{B_e \rightarrow 0} \text{var}\left(x_{\text{CoP}}(t; f, B_e)\right) = G_{x_{\text{CoP}} x_{\text{CoP}}}(f).$$

Thus, the frequency-domain intersection-point height measure and its goodness-of-fit can be computed from the spectral densities as follows:

$$z_{\text{IP,LR}}(f) = \frac{\Re\left\{G_{\theta_F x_{\text{CoP}}}(f)\right\}}{G_{\theta_F \theta_F}(f)}, \quad (13)$$

$$R^2(f) = \frac{\Re\left\{G_{\theta_F x_{\text{CoP}}}(f)\right\}^2}{G_{\theta_F \theta_F}(f)G_{x_{\text{CoP}} x_{\text{CoP}}}(f)}. \quad (14)$$

Note that the one-sided spectral densities can be computed from the Fourier Transforms of θ_F and x_{CoP} :

$$G_{\theta_F x_{\text{CoP}}}(f) = \lim_{T_s \rightarrow \infty} \frac{2}{T_s} E\left[\theta_F(f; T_s)^* x_{\text{CoP}}(f; T_s)\right],$$

$$G_{\theta_F \theta_F}(f) = \lim_{T_s \rightarrow \infty} \frac{2}{T_s} E\left[\theta_F(f; T_s)^* \theta_F(f; T_s)\right].$$

where $\theta_F(f; T_s)$ and $x_{\text{CoP}}(f; T_s)$ denote the Fourier Transforms of $\theta_F(t)$ and $x_{\text{CoP}}(t)$, T_s denotes the sample length of the time-series data, $\cdot^*$ denotes the complex conjugate operator, and $E[\cdot]$ denotes the expectation operator.

3.2 Closed-form expression of model intersection-point height

Sugimoto-Dimitrova et al. showed that the original frequency-domain intersection-point-height measure can be derived exactly for any linear model of human standing that is formulated with internal biological noise as the system inputs and θ_F and x_{CoP} as its outputs [8]. Likewise, the new LR-based intersection-point height can also be derived for such a linear model.

Consider the frequency-response-function representation of the linear human-standing model:

$$\mathbf{y}(f) = \mathbf{H}(f)\mathbf{w}(f),$$

where $\mathbf{y} = [\theta_F, x_{\text{CoP}}]^T$ is the output vector of foot-ground force measurements ($\cdot^T$ is the matrix transpose operator), $\mathbf{w}$ is the input vector of biological noise processes, and $\mathbf{H}(f)$ is the frequency response function relating the two. The output spectral density matrix $\mathbf{G}_{\mathbf{yy}}(f)$ can be estimated from the frequency response function $\mathbf{H}(f)$ and the input spectral density matrix $\mathbf{G}_{\mathbf{ww}}(f)$:

$$\mathbf{G}_{\mathbf{yy}}(f) = \mathbf{H}(-f)\mathbf{G}_{\mathbf{ww}}(f)\mathbf{H}(f)^T,$$

where

$$\mathbf{G}_{\mathbf{ww}}(f) = \lim_{T_s \rightarrow \infty} \frac{2}{T_s} E \left[\mathbf{w}(-f; T_s)^T \mathbf{w}(f; T_s) \right].$$

This output spectral density is composed of the one-sided auto- and cross-spectral densities:

$$\mathbf{G}_{\mathbf{yy}}(f) = \begin{bmatrix} G_{\theta_F \theta_F}(f) & G_{\theta_F x_{\text{CoP}}}(f) \\ G_{x_{\text{CoP}} \theta_F}(f) & G_{x_{\text{CoP}} x_{\text{CoP}}}(f) \end{bmatrix},$$

which can be used to obtain the frequency-domain LR-based intersection-point height for the model along with its coefficient of determination from 13 and 14.

MATLAB code for the LR-based intersection-point-height analysis is available at https://github.com/rikasd/zIP_poststroke. The code base provides an example application of the method to investigate balance control in survivors of stroke [6], and includes code for processing raw foot-ground-interaction-force data as well as for finding the best-fit computational models describing the observed balance behavior.

4 Discussion

In this work, a geometrically-informed modification to the intersection-point-height measure was presented to capture the closed-loop dynamics of human standing balance.

The revised intersection-point-height measure preserves the useful qualities of the original measure. As with the original measure, the frequency-dependent linear-regression-based intersection-point height is related to the closed-loop dynamics, capturing not only aspects of biomechanics but also of neural control. The measure remains insensitive to the specific noise dynamics (e.g. white noise versus low-pass-filtered white noise), allowing the researcher to focus on other aspects of the system dynamics, such as the apparent controller, without worrying about the exact nature of noise [8].

The revised measure still shares many of the limitations of the original method. Firstly, the geometrically-derived measure assumed that the vertical force dominated over the shear force ($F_z \gg F_x$) in order to arrive at the simplified form equivalent to linear regression. However, this assumption is generally valid for quiet standing and does not cause much deviation from the original geometric measure. Secondly, the measure assumes stationarity of the time-series data, while experimental human standing data exhibit apparent non-stationarity [10]. However, because measurements collected are finite, with proper data processing, such as window tapering and zero-padding before filtering, one can obtain adequate estimates of the data spectrum. Sugimoto-Dimitrova et al. also showed that, despite the apparent non-stationarity of biological data, variations in the window size used in computing the spectral densities produced consistent IP height estimates at higher frequencies, giving confidence in the measure [8]. Third, the modeling aspect requires a linear descriptive model to arrive at an exact closed-form expression of the model's IP height. Although in principal the biological system is nonlinear, a linear approximation may still be good enough, particularly for

quiet standing where the motions are small and the dynamics are slow. Furthermore, if one were to develop a nonlinear model, it is still possible to obtain an estimate of the IP height using simulations that output the foot-force measurements and processing those just like experimental data. Finally, as with the original intersection-point-height measure, the updated measure does not capture full system dynamics: for instance, the measure is insensitive to the specific noise dynamics. Although this insensitivity to certain aspects of the system dynamics has its advantages (e.g. not having to worry about modeling the exact nature of biological noise), it would be useful to look at other measures in parallel with the IP height to get a more complete picture of the balance dynamics.

5 Conclusion

The intersection-point-height measure of human standing has been widely applied to various populations engaged in assorted unperturbed standing balance tasks. The significant differences in the IP-height measure observed with age and impairment, along with its close relation to the closed-loop balance dynamics, highlight the measure’s potential to be used in studying the balance mechanisms in healthy versus impaired individuals. These features also make it a promising measure for being used in the clinical setting as a diagnostic tool, to identify how the balance controller has been compromised, and to inform therapy. The IP-height measure might also be relevant in exo-skeletal devices used to augment human performance in a manufacturing or military context, as it provides a means to monitor the balance dynamics of the combined human-exo system to ensure that these devices provide augmentation without disrupting posture control. Overall, the IP-height measure shows potential as a tool to help advance human balance research and the development of therapeutic and assistive devices.

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